

One Zone Disproves the Printed Zone–Inradius Conjecture

Run `fa_banach.001`, lane 2

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Status. Candidate counterexample, likely valid, for human review.

Outcome. Conjecture 8 in Polyanskii’s *A cap covering theorem* is false as literally printed. In every dimension, any single nondegenerate zone is already a counterexample: the conjectured left-hand side equals π , whereas the claimed lower bound is 2π .

1 Source question

Let $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ be the unit sphere. The source defines a zone of width 2α as the set of points at spherical distance at most α from a great sphere [1, p. 2]. Its final open problem is the following.

We finish with the following conjecture generalizing Theorems 3 and 5 posed by Maxim Didid.

Conjecture 8. *Let $Z_1, \dots, Z_n \subset S$ be zones of width $\beta_1, \dots, \beta_n$, respectively. If $S \setminus \bigcup_{i=1}^n Z_i$ consists of convex connected components $L_1, \dots, L_{2m}$ with inradius $\gamma_1, \dots, \gamma_{2m}$, respectively, then $\beta_1 + \dots + \beta_n + \gamma_1 + \dots + \gamma_{2m} \geq 2\pi$.*

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$$\beta_1 + \dots + \beta_n + \gamma_1 + \dots + \gamma_{2m} \geq 2\pi.$$

The paper attributes the conjecture to “Maxim Didid”; its source label is `conjecture:didid`. We refer to it simply as the printed zone–inradius conjecture.

2 Claimed contribution

Theorem 1. *Fix $d \geq 1$ and $0 < \beta < \pi$. Let $Z \subset \mathbb{S}^d$ be any zone of width β . Then $\mathbb{S}^d \setminus Z$ has exactly two convex connected components, each of spherical inradius*

$$\gamma = \frac{\pi - \beta}{2}.$$

Consequently, with $n = 1$ and $m = 1$, the left-hand side of Conjecture 8 is

$$\beta + \gamma + \gamma = \pi < 2\pi.$$

Thus Conjecture 8 is false as stated.

3 Proof intuition

A zone is a symmetric belt around an equator. If its half-width is $\beta/2$, then each of the two polar caps left over has angular radius $\pi/2 - \beta/2$. The belt width plus the radii of both caps therefore telescopes:

$$\beta + 2 \left(\frac{\pi}{2} - \frac{\beta}{2} \right) = \pi.$$

This is independent of both the width and the dimension, and is only half of the conjectured lower bound.

4 Proof

Choose a unit vector $u \in \mathbb{R}^{d+1}$ normal to the great sphere around which Z is centered. Put $a = \beta/2$. For $x \in \mathbb{S}^d$, the spherical distance from x to the great sphere $u^\perp \cap \mathbb{S}^d$ is

$$\arcsin |\langle x, u \rangle|.$$

Since the source's zones are closed, it follows that

$$Z = \{x \in \mathbb{S}^d : |\langle x, u \rangle| \leq \sin a\}.$$

Hence

$$\mathbb{S}^d \setminus Z = L_+ \dot{\cup} L_-, \quad L_\pm = \{x \in \mathbb{S}^d : \pm \langle x, u \rangle > \sin a\}.$$

Because

$$\sin a = \cos \left(\frac{\pi}{2} - a \right),$$

and $\text{dist}_{\mathbb{S}^d}(x, u) = \arccos \langle x, u \rangle$, we have

$$L_+ = \left\{ x \in \mathbb{S}^d : \text{dist}_{\mathbb{S}^d}(x, u) < \frac{\pi}{2} - a \right\},$$

with the analogous formula for L_- centered at $-u$. Thus L_+ and L_- are antipodal open spherical caps of radius

$$r = \frac{\pi}{2} - a = \frac{\pi - \beta}{2}.$$

Since $0 < r < \pi/2$, each cap is geodesically convex and connected, and there are no other components.

For completeness, the inradius of either open cap is exactly r . Every concentric cap of radius $r' < r$ is contained in it, so its inradius is at least r . Conversely, if a spherical cap of radius ρ is contained in L_+ and has center y , the triangle inequality along a minimizing geodesic from u through y gives

$$\rho \leq r - \text{dist}_{\mathbb{S}^d}(u, y) \leq r.$$

Equivalently, if inradius is defined as a supremum for open domains, the supremum is r . Therefore $\gamma_1 = \gamma_2 = r$, and the quantity in the conjecture is

$$\beta + \gamma_1 + \gamma_2 = \beta + 2 \left(\frac{\pi - \beta}{2} \right) = \pi < 2\pi.$$

All of the hypotheses of Conjecture 8 hold with $n = 1$ and $2m = 2$, completing the proof.

5 Hypothesis audit

Printed requirement	One-zone construction	Check
Zones $Z_1, \dots, Z_n$	one zone Z , $n = 1$	yes
Widths β_i	$\beta_1 = \beta \in (0, \pi)$	yes
Even number of components	L_+, L_- , so $2m = 2$	yes
Components connected	open caps	yes
Components convex	radii $< \pi/2$	yes
Component inradii	$\gamma_1 = \gamma_2 = (\pi - \beta)/2$	yes
Claimed inequality	$\pi \geq 2\pi$	false

6 Robustness, interpretation, and limitations

The counterexample does not depend on a limiting or degenerate zone: every $\beta \in (0, \pi)$ works. For example, $\beta = \pi/2$ leaves two caps of inradius $\pi/4$, and the conjectured sum is again π .

The open/closed convention does not change the calculation. The source defines the zone as the closure of the intersection with an open plank, so its complementary components are open. Their inradius, in the standard supremal sense for domains, is $(\pi - \beta)/2$. Passing to closures yields closed caps with the same numerical inradius.

The factor-of-two discrepancy strongly suggests that the displayed constant, one of the summands, or a normalization was misstated. This packet disproves only the literal printed statement; it does not assert which correction was intended, and it does not address Conjecture 7 in the same paper.

7 Verification and novelty search

The script `code/verify_one_zone.py` checks the exact symbolic identity and several sample widths. It runs with:

```
conda run --no-capture-output -n sandbox python \  
code/verify_one_zone.py
```

Before promotion, the run’s cheap title, arXiv-id, and keyword indexes were searched. Targeted web searches for the exact displayed inequality, the paper title together with “correction” and “erratum,” and the zone/inradius keywords found no later correction or resolution as of 23 July 2026. The local arXiv source and PDF both contain the 2π constant. This is a bounded novelty check, not a comprehensive literature review.

8 Human review recommendation

A reviewer in discrete or spherical geometry should first confirm the source’s intended convention for “inradius” of an open component and ask the author whether 2π is a typographical or normalization error. Under the definitions printed in the paper, the proof above is complete.

References

- [1] A. Polyanskii, *A cap covering theorem*, *Combinatorica* **41** (2021), 695–702; arXiv:2006.02192.